

求一个4次多项式 $p(x)$, 使得 $p(-1) = 1, p(0) = 1, p(1) = 5, p(2) = 13, p(-2) = 29$

1. 解:

$$\text{令 } w(x) = (x+2)(x+1)x(x-1)(x-2)$$

$$\text{则 } w'(-2) = 24 \quad w'(-1) = -6 \quad w'(0) = 4 \quad w'(1) = -6 \quad w'(2) = 24$$

$$b_0(x) = \frac{w(x)}{(x-x_0)w'(-2)} = \frac{1}{24}x(x+1)(x-1)(x-2) = \frac{1}{24}(x^4 - 2x^3 - x^2 + 2x)$$

$$b_1(x) = \frac{w(x)}{(x-x_1)w'(-1)} = -\frac{1}{6}x(x+2)(x-1)(x-2) = -\frac{1}{6}(x^4 - x^3 - 4x^2 + 4x)$$

$$b_2(x) = \frac{w(x)}{(x-x_2)w'(0)} = \frac{1}{4}(x+2)(x+1)(x-1)(x-2) = \frac{1}{4}(x^4 - 5x^2 + 4)$$

$$b_3(x) = \frac{w(x)}{(x-x_3)w'(1)} = -\frac{1}{6}x(x+2)(x+1)(x-2) = -\frac{1}{6}(x^4 + x^3 - 4x^2 - 4x)$$

$$b_4(x) = \frac{w(x)}{(x-x_4)w'(2)} = \frac{1}{24}x(x+2)x(x+1)(x-1) = \frac{1}{24}(x^4 + 2x^3 - x^2 - 2x)$$

$$\therefore p(x) = y_0 b_0(x) + y_1 b_1(x) + y_2 b_2(x) + y_3 b_3(x) + y_4 b_4(x)$$

$$= \left(\frac{29}{24} - \frac{1}{6} + \frac{1}{4} - \frac{5}{6} + \frac{13}{24}\right)x^4 + \left(-\frac{29}{12} + \frac{1}{6} - \frac{5}{6} + \frac{13}{12}\right)x^3 + \left(-\frac{29}{24} + \frac{2}{3} - \frac{5}{4} + \frac{10}{3} - \frac{13}{24}\right)x^2 +$$

$$\left(\frac{29}{12} - \frac{2}{3} + \frac{10}{3} - \frac{13}{12}\right)x + 1$$

$$= x^4 - 2x^3 + x^2 + 4x + 1$$

解法二: Newton插值多项式

x	$f(x)$	一阶均差	二阶均差	三阶均差	四阶均差
-2	29	$\frac{1-29}{-1+2} = 28$	$\frac{0+28}{0+2} = 14$	$\frac{2-14}{1+2} = -4$	$\frac{0+4}{2+2} = 1$
-1	1	$\frac{1-1}{0+1} = 0$	$\frac{4-0}{1+1} = 2$	$\frac{2-2}{2+1} = 0$	
0	1	$\frac{5-1}{1-0} = 4$	$\frac{8-4}{2-0} = 2$		
1	5	$\frac{13-5}{2-1} = 8$			
2	13				

$$p_4(x) = 29 - 28(x+2) + 14(x+2)(x+1) - 4(x+2)(x+1)x + (x+2)(x+1)x(x-1)$$

$$= x^4 - 2x^3 + x^2 + 4x + 1$$

☆ 设 $x_0, x_1, \dots, x_n$ 是互异的插值节点组, $l_i(x)$ 是 Lagrange 插值基函数.

$$l_i(x) = \frac{\omega_{n+1}(x)}{(x-x_i)\omega'_{n+1}(x_i)}, \quad \omega_{n+1}(x) = (x-x_0)\cdots(x-x_n).$$

试求 $\sum_{i=0}^n x_i^k l_i(x), 0 \leq k \leq n$.

2. 解:

设原函数 $f(x) = x^k, 0 \leq k \leq n$, 且其 Lagrange 插值为 $p_n(x)$, 则有

$$y_0 = f(x_0) = x_0^k, \quad y_1 = f(x_1) = x_1^k, \quad \dots \quad y_n = f(x_n) = x_n^k$$

$$p_n(x) = \sum_{i=0}^n y_i l_i(x) = \sum_{i=0}^n x_i^k l_i(x)$$

$$\text{余项 } r_n(x) = f(x) - p_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} \omega_{n+1}(x)$$

当 n 足够大时, 认为 $r_n(x) = 0 \Rightarrow p_n(x) = f(x)$

$$\text{则有 } \sum_{i=0}^n x_i^k l_i(x) = x^k$$

求 $[0,1]$ 上以 $\rho(x) = 1$ 为权函数的标准正交多项式系 $\psi_0(x), \psi_1(x), \psi_2(x)$.

3. 解:

设首项 $\phi_0(x) = 1, \varphi_0(x) = 1, \varphi_1(x) = x, \varphi_2(x) = x^2$

$$\phi_1(x) = \begin{vmatrix} (\varphi_0, \varphi_0) & \varphi_0 \\ (\varphi_1, \varphi_0) & \varphi_1 \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ \int_0^1 x dx & x \end{vmatrix} = x - \frac{1}{2}$$

$$\begin{aligned} \phi_2(x) &= \begin{vmatrix} (\varphi_0, \varphi_0) & (\varphi_0, \varphi_1) & \varphi_0 \\ (\varphi_1, \varphi_0) & (\varphi_1, \varphi_1) & \varphi_1 \\ (\varphi_2, \varphi_0) & (\varphi_2, \varphi_1) & \varphi_2 \end{vmatrix} = \begin{vmatrix} 1 & \int_0^1 x dx & 1 \\ \int_0^1 x dx & \int_0^1 x^2 dx & x \\ \int_0^1 x^2 dx & \int_0^1 x^3 dx & x^2 \end{vmatrix} \\ &= \begin{vmatrix} 1 & \frac{1}{2} & 1 \\ \frac{1}{2} & \frac{1}{3} & x \\ \frac{1}{3} & \frac{1}{4} & x^2 \end{vmatrix} = \frac{1}{3}x^3 - \frac{1}{4}x - \frac{1}{2}(\frac{1}{2}x^2 - \frac{1}{3}x) + \frac{1}{8} - \frac{1}{9} \\ &= \frac{1}{12}x^3 - \frac{1}{12}x + \frac{1}{12} \end{aligned}$$

$$\Delta_0 = (\phi_0, \varphi_0) = 1 \quad \Delta_1 = (\phi_1, \varphi_1) = \int_0^1 x^2 - \frac{1}{2}x dx = \frac{1}{12}$$

$$\Delta_2 = (\phi_2, \varphi_2) = \int_0^1 \frac{1}{12}x^6 - \frac{1}{12}x^3 + \frac{1}{12}x^4 dx = \frac{1}{60} - \frac{1}{48} + \frac{1}{216} = \frac{1}{2160}$$

$$\Rightarrow \varphi_0(x) = 1$$

$$\varphi_1(x) = \sqrt{12x} (x - \frac{1}{2}) = 2\sqrt{3}x - \sqrt{3}$$

$$\varphi_2(x) = \sqrt{12x \cdot 2160} (\frac{1}{12}x^3 - \frac{1}{12}x + \frac{1}{12}) = 6\sqrt{5}x^2 - 6\sqrt{5}x + \sqrt{5}$$

★用最小二乘法求拟合下列数据的二次多项式： ax^2+bx+c

x_i	1	3	4	5	6	7	8	9	10
y_i	10	5	4	2	1	1	2	3	4

4. 解=

$$\text{取 } \varphi_0(x)=1 \quad \varphi_1(x)=x \quad \varphi_2(x)=x^2$$

$$\text{则 } (\varphi_0, \varphi_0)=9 \quad (\varphi_0, \varphi_1)=53 \quad (\varphi_0, \varphi_2)=381$$

$$(\varphi_1, \varphi_0)=53 \quad (\varphi_1, \varphi_1)=381 \quad (\varphi_1, \varphi_2)=3017$$

$$(\varphi_2, \varphi_0)=381 \quad (\varphi_2, \varphi_1)=3017 \quad (\varphi_2, \varphi_2)=25317$$

$$(f, \varphi_0)=32 \quad (f, \varphi_1)=147 \quad (f, \varphi_2)=1025$$

$$\begin{bmatrix} 9 & 53 & 381 \\ 53 & 381 & 3017 \\ 381 & 3017 & 25317 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 32 \\ 147 \\ 1025 \end{bmatrix} \Rightarrow \begin{bmatrix} \sum_{i=1}^n 1 & \sum_{i=1}^n x_i & \sum_{i=1}^n x_i^2 \\ \sum_{i=1}^n x_i & \sum_{i=1}^n x_i^2 & \sum_{i=1}^n x_i^3 \\ \sum_{i=1}^n x_i^2 & \sum_{i=1}^n x_i^3 & \sum_{i=1}^n x_i^4 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^n y_i \\ \sum_{i=1}^n x_i y_i \\ \sum_{i=1}^n x_i^2 y_i \end{bmatrix}$$

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$$\Rightarrow a_0=13.4597, a_1=-3.6053, a_2=0.2676$$

$$\therefore \text{二次多项式为 } 0.2676x^2 - 3.6053x + 13.4597$$

判断下列函数是否为各自定义域上的三次样条函数：

$$(1) s(x) = \begin{cases} 2 + 3x + 4x^2 + x^3, & x \in [-1, 0]; \\ 2 + 3x + 4x^2 + 4x^3, & x \in [0, 1]; \\ 1 + 6x + x^2 + 5x^3, & x \in [1, 2]. \end{cases}$$

$$(2) s(x) = \begin{cases} x^3 + 2x + 1, & x \in [0, 1]; \\ x^3 + x^2 + 2, & x \in [1, 2]. \end{cases}$$

5. 解

$$(1) s(x) = \begin{cases} 2 + 3x + 4x^2 + x^3, & x \in [-1, 0] \\ 2 + 3x + 4x^2 + 4x^3, & x \in [0, 1] \\ 1 + 6x + x^2 + 5x^3, & x \in [1, 2] \end{cases}$$

$$p_1(x) - p_0(x) = 3x^3$$

$$p_2(x) - p_1(x) = x^3 - 3x^2 + 3x - 1 = (x-1)^3$$

∴ 该函数是三次样条函数

$$(2) s(x) = \begin{cases} x^3 + 2x + 1, & x \in [0, 1] \\ x^3 + x^2 + 2, & x \in [1, 2] \end{cases}$$

$$p_1(x) - p_0(x) = x^3 - 2x + 1 = (x-1)^2 \neq (x-1)^3$$

∴ 该函数不是三次样条函数

根据如下离散数据, 请用最小二乘法拟合出形如 $y = \frac{x}{ax+b}$ 的曲线。

x_i	1	1/2	1/3	1/4	1/5
y_i	1/2	1/4	5/32	1/8	5/43

6. 解:

$$\text{由 } y = \frac{x}{ax+b} \quad \therefore \frac{1}{y} = \frac{ax+b}{x} = a + b \frac{1}{x}$$

$x' = \frac{1}{x}$	1	2	3	4	5
$y' = \frac{1}{y}$	2	4	$\frac{32}{5}$	8	$\frac{42}{5}$

$$\sum_{i=1}^5 x' = 15 \quad \sum_{i=1}^5 x'^2 = 55$$

$$\sum_{i=1}^5 y' = 29 \quad \sum_{i=1}^5 x' \cdot y' = 104.2$$

$$\Rightarrow \begin{pmatrix} 5 & 15 \\ 15 & 55 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 29 \\ 104.2 \end{pmatrix} \Rightarrow \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0.64 \\ 1.72 \end{pmatrix}$$

$$\therefore y = \frac{x}{0.64x + 1.72}$$

确定下列函数组在指定的区间上是否满足 Haar 条件

(1) $\{1, x, x^2, \dots, x^m\}$ 在任意闭区间 $[a, b]$ 上;

(2) $\{1, x^2, x^4, x^6\}$ 在任意闭区间 $[0, 1]$ 上;

(3) $\{1, x^2, x^4\}$ 在任意闭区间 $[-1, 1]$ 上。

7. 解 =

(1) $\{1, x, x^2, \dots, x^m\}$ 在任意闭区间 $[a, b]$ 上;

$$\begin{vmatrix} 1 & x_1 & \dots & x_1^m \\ 1 & x_2 & \dots & x_2^m \\ \vdots & \vdots & \dots & \vdots \\ 1 & x_m & \dots & x_m^m \end{vmatrix} = \prod_{1 \leq i < j \leq m} (x_j - x_i)$$

$\because x_1, \dots, x_m$ 互异, $\therefore \prod_{1 \leq i < j \leq m} (x_j - x_i) \neq 0$ $\therefore$ 满足 Haar 条件

(2) $\{1, x^2, x^4, x^6\}$ 在任意闭区间 $[0, 1]$ 上

$$\begin{vmatrix} 1 & x_1^2 & x_1^4 & x_1^6 \\ 1 & x_2^2 & x_2^4 & x_2^6 \\ 1 & x_3^2 & x_3^4 & x_3^6 \\ 1 & x_4^2 & x_4^4 & x_4^6 \end{vmatrix} = \prod_{1 \leq i < j \leq 4} (x_j^2 - x_i^2)$$

$\because x_1, \dots, x_4$ 互异, 且在 $[0, 1]$ 上, $\therefore x_j^2 \neq x_i^2$, 行列式不为 0 $\therefore$ 满足 Haar 条件

(3) $\{1, x^2, x^4\}$ 在任意闭区间 $[-1, 1]$

$$\begin{vmatrix} 1 & x_1^2 & x_1^4 \\ 1 & x_2^2 & x_2^4 \\ 1 & x_3^2 & x_3^4 \end{vmatrix} = (x_1^2 - x_2^2)(x_3^2 - x_1^2)(x_3^2 - x_2^2)$$

$\because x_1, x_2, x_3$ 在 $[-1, 1]$ 上 $\therefore x_1, x_2, x_3$ 中可能存在 $x_i^2 = x_j^2$, 行列式不但为 0 $\therefore$ 不满足 Haar 条件